Exercise 15.5

Solve the following equations.

1. $\frac{dy}{dx} + 3y = e^{-x}$

Sol^n: $\frac{dy}{dx} + 3y = e^{-x}$ ..... (i)

Which is linear differential equation of the form $\frac{dy}{dx} + Py = Q$, where $P = 3$, $Q = e^{-x}$

$$I.F. = e^{\int P dx} = e^{\int 3.dx} = e^{3x}$$

$\therefore$ The solution of the equation is

$$y \times I.F. = \int (Q \times I.F.) dx$$

i.e. $y.e^{3x} = \int e^{2x}$ $\quad\quad$ $\left[\because \text{i.e. } \int y \,.\, (If) = \int e^{2x}\right]$

$\Rightarrow$ $y.e^{3x} = \frac{e^{2x}}{2} + c$

$\therefore$ $y = \frac{1}{2} e^{-x} + c\, e^{-3x}$.

2. $\frac{dy}{dx} + \frac{3y}{x} = 9x^5$

Sol^n: $\frac{dy}{dx} + \frac{3y}{x} = 9x^5$ ..... (i)

Whichislinear differential equation of the form $\frac{dy}{dx} + Py = Q$, where $P = \frac{3}{x}$, $Q = 9x^5$

$$I.F. = e^{\int P dx} = e^{\int \frac{3}{x}} = e^{3\log x} = e^{\log x^3} = x^3$$

$\therefore$ The solution of the equation is

$$y \times \text{I.F.} = \int (Q \times I.F.)\, dx$$

$\Rightarrow \quad yx^3 = \int 9x^5 \,.\, x^3 \,.\, dx$

$\Rightarrow \quad yx^3 = 9\frac{x^9}{9} + c$

$\therefore \quad y = x^6 + \frac{c}{x^3}$.

3. $\mathbf{x\frac{dy}{dx} - y = 2x^3}$

Sol^n: $\frac{dy}{dx} - \frac{y}{x} = 2x^2$ ..... (i)

Whichis linear differential equation of the form $\frac{dy}{dx} + Py = Q$, where $P = \frac{-1}{x}$, $Q = 2x^2$

$\therefore \quad \text{I.F.} = e^{\int P dx} = e^{\int \frac{-1}{x} dx} = e^{-\log x} = x^{-1}$

$\therefore$ The solution of the equation is

$$y \times \text{I.F.} = \int (Q \times I.F.)\, dx$$

$\Rightarrow \quad y \,.\, x^{-1} = \int 2x.dx$

$\Rightarrow \quad y \,.\, x^{-1} = \frac{2x^2}{2} + c$

$\therefore \quad y = x^3 + cx.$

4. $\mathbf{(1 + x^2)\frac{dy}{dx} + 2xy = 4x^2}$

Sol^n: $\frac{dy}{dx} + \frac{2xy}{1 + x^2} = \frac{4x^2}{(1 + x^2)}$ ..... (i)

Whichis linear differential equation of the form $\frac{dy}{dx} + Py = Q$, where $P = \frac{2x}{1 + x^2}$, $Q = \frac{4x^2}{(1 + x^2)}$

$$\text{I.F.} = e^{\int P dx} = e^{\int \frac{2x}{1 + x^2}} = e^{\log(1 + x^2)} = (1 + x^2)$$

$\therefore$ The solution of the equation is

$$y \times \text{I.F.} = \int (Q \times I.F.)\, dx$$

$$(1 + x^2)\, y = \int 4x^2 dx$$

$\Rightarrow \quad (1 + x^2)\, y = \frac{4x^3}{3} + c$

$\therefore \quad (1 + x^2)\, y = \frac{4}{3}x^3 + c.$

5. $(x+1)\frac{dy}{dx} + 2y = \frac{e^x}{x+1}$

Sol^n: $\frac{dy}{dx} + \frac{2}{(x+1)}y = \frac{e^x}{(1+x)^2}$ ..... (i)

Whichis linear differential equation of the form $\frac{dy}{dx} + Py = Q$, where $P = \frac{2}{x+1}$, $Q = \frac{e^x}{(1+x)^2}$

$I.F = e^{\int P dx} = e^{\int \frac{2}{x+1}} = e^{2\log(x+1)} = (x+1)^2$

$\therefore$ The solution of the equation is

$y \times \text{I.F.} = \int (Q \times I.F.)\, dx$

$\Rightarrow \quad y(x+1)^2 = \int e^x dx$

$\therefore \quad y(x+1)^2 = e^x + c.$

6. $\sin x \frac{dy}{dx} + \cos x \; y = \sin x \cos x$

Sol^n: $\frac{dy}{dx} + \frac{\cos x}{\sin x}y = \cos x$ ..... (i)

Which is linear differential equation of the form $\frac{dy}{dx} + Py = Q$, where $P = \frac{\cos x}{\sin x}$ and $Q = \sin x$

$I.F. = e^{\int P dx} = e^{\int \frac{\cos x}{\sin x} dx} = e^{\log \sin x} = \sin x$

$\therefore$ The solution of the equation is

$y \times \text{I.F.} = \int (Q \times I.F.)\, dx$

Integrating, we get

$y \sin x = \int \cos x \sin x \, dx$

$\Rightarrow \quad y \sin x = \frac{1}{2}\int \sin 2x \, dx$

$\Rightarrow \quad y \sin x = -\frac{1}{2}\frac{\cos 2x}{2} + c$

$\Rightarrow \quad y \sin x + \frac{1}{4}\cos 2x = c.$

7. $\tan x \frac{dy}{dx} + y = \sec x$

Sol^n: $\frac{dy}{dx} + \frac{1}{\tan x}y = \frac{\sec x}{\tan x}$

$\Rightarrow \quad \frac{dy}{dx} + \cot x.\, y = \text{cosec } x$ ..... (i)

Whichis linear differential equation of the form $\frac{dy}{dx} + Py = Q$,where, $P = \cot x$, $Q = \text{cosec } x$

$I.F. = e^{\int P dx} = e^{\int \frac{\cos x}{\sin x} dx} = e^{\log \sin x} = \sin x$

$\therefore$ The solution of the equation is

$$y \times \text{I.F.} = \int (Q \times I.F.)\, dx$$

$\Rightarrow$ $y \cdot \sin x = \int dx$

$\Rightarrow$ $y \sin x = x + c$

8. $\mathbf{\frac{dy}{dx} + 2y \tan x \cdot y = \sin x}$

Sol^n: $\frac{dy}{dx} + 2 \tan x \cdot y = \sin x.$

Whichislinear differential equation of the form $\frac{dy}{dx} + Py = Q$, whereP = 2tan x, Q = sin x

$$\text{I.F.} = e^{\int P dx} = e^{2\int \frac{\sin x}{\cos x} dx} = e^{-2\log \sin x} = e^{-2\log \cos x} = e^{\log(1/\cos x)^2} = e^{\log(\sec^2 x)} = \sec^2 x$$

$\therefore$ The solution of the equation is

$$y \times \text{I.F.} = \int (Q \times I.F.)\, dx$$

$\Rightarrow$ $y.\sec^2 x = \int \sec x \tan x \, dx$

$\Rightarrow$ $y.\sec^2 x = \sec x + c$

$\Rightarrow$ $y = \cos x + c \cos^2 x.$

9. $\mathbf{(1 - x^2)\frac{dy}{dx} - xy = 1}$

Sol^n: $\frac{dy}{dx} - \frac{x}{1 - x^2} y = \frac{1}{1 - x^2}$ ..... (i)

Whichis linear differential equation of the form $\frac{dy}{dx} + Py = Q$,Here, $P = -\frac{x}{1 - x^2}$, $Q = \frac{1}{1 - x^2}$

$$\text{I.F.} = e^{\int \frac{-x}{1 - x^2} dx} = e^{\frac{1}{2}\log(1 - x^2)} = e^{\log \sqrt{1 - x^2}} = \sqrt{1 - x^2}$$

$\therefore$ The solution of the equation is

$$y \times \text{I.F.} = \int (Q \times I.F.)\, dx$$

$\Rightarrow$ $y\sqrt{1 - x^2} = \int \frac{1}{\sqrt{1 - x^2}} dx$

$\Rightarrow$ $y\sqrt{1 - x^2} = \sin^{-1} x + c$

10. $\mathbf{\frac{dy}{dx} + \frac{2x}{1 + x^2} y = \frac{1}{(1 + x^2)^2}}$

Sol^n: $\frac{dy}{dx} + \frac{2x}{1 + x^2} y = \frac{1}{(1 + x^2)^2}$ ... (i)

Whichis linear differential equation of the form $\frac{dy}{dx} + Py = Q$, where$P = \frac{2x}{1 + x^2}$, $Q = \frac{1}{(1 + x^2)^2}$

$$\text{I.F.} = e^{\int P dx} = e^{\log(1 + x^2)} = 1 + x^2$$

$\therefore$ The solution of the equation is

$$y \times \text{I.F.} = \int (Q \times I.F.)\, dx$$

$\Rightarrow \quad y(1 + x^2) = \int \frac{1}{1 + x^2}\, dx \qquad \Rightarrow \qquad y(1 + x^2) = \tan^{-1} x + c.$

11. $\mathbf{\frac{dy}{dx} - 2xy = x}$

Sol^n: $\frac{dy}{dx} - 2xy = x$ ..... (i)

This is linear differential equation $\frac{dy}{dx} + Py = Q$, where $P = -2x$, $Q = x$

$$\text{I.F.} = e^{\int P dx} = e^{\int -2x\, dx} = e^{-x^2}$$

$\therefore$ The solution of the equation is

$$y \times \text{I.F.} = \int (Q \times I.F.)\, dx$$

$\Rightarrow \quad ye^{-x^2} = \frac{-1}{2} e^{-x^2} + c \qquad \Rightarrow y = -\frac{1}{2} + ce^{-x^2}$

12. $\mathbf{\frac{dy}{dx} + \frac{y}{x^2} = \frac{1}{x^2}}$

Sol^n: $\frac{dy}{dx} + \left(\frac{1}{x^2}\right) y = \frac{1}{x^2}$ ..... (i)

Whichis linear differential equation of the form $\frac{dy}{dx} + Py = Q$, where $P = \frac{1}{x^2}$, $Q = \frac{1}{x^2}$

$$\text{I.F.} = e^{\int P dx} = e^{\int \frac{1}{x^2} dx} = e^{-1/x}$$

$\therefore$ The solution of the equation is

$$y \times \text{I.F.} = \int (Q \times I.F.)\, dx$$

$\Rightarrow \quad ye^{-1/x} = \int \frac{1}{x^2} e^{-1/x}\, dx$

$\Rightarrow \quad ye^{-1/x} = e^{-1/x} + c \qquad \Rightarrow y = 1 + ce^{1/x}.$

13. $\mathbf{(1 + x)\frac{dy}{dx} - xy = 1 - x}$

Sol^n: $\frac{dy}{dx} + \frac{(-x)}{1 + x} y = \frac{1 - x}{1 + x}$ ..... (i)

Whichis linear differential equation of the form $\frac{dy}{dx} + Py = Q$,where $P = \frac{-x}{1 + x}$, $Q = \frac{1 - x}{1 + x}$

Now, $\int P dx = \int \frac{-x}{1 + x} dx = \int \frac{[1 - (1 + x)]}{1 + x} = \log (1 + x) - x$

$$\text{I.F.} = e^{\int P dx} = e^{\log(1 + x) - x} = e^{\log(1 + x)} \cdot e^{-x} = (1 + x) e^{-x}$$

$\therefore$ The solution of the equation is

$$y \times \text{I.F.} = \int (Q \times I.F.)\, dx$$

$\Rightarrow$ $(1 + x)e^{-x}y = \int (1 - x)\, e^{-x}\, dx$

$= (1 - x) \cdot \left(\frac{e^{-x}}{-1}\right) - \int - e^{-x} \cdot (-1)\, dx = -(1 - x)\, e^{-x} - \int e^{-x}\, dx$

$= [-(1 - x)\, e^{-x} + e^{-x}] + c = e^{-x}\, (-1 + x + 1) + c$

$\Rightarrow$ $(1 + x)e^{-x}y = ex^{-x} + c$

$\therefore$ $(1 + x)\, y = x + ce^{x}$.

14. $\mathbf{x \frac{dy}{dx} + 2y = x^2 \log x}$

Sol^n: $\frac{dy}{dx} + \left(\frac{2}{x}\right) y = x \log x$ ..... (i)

This equation is linear differential equation of the form $\frac{dy}{dx} + Py = Q$, where

$P = 2/x$ and $Q = x \log x$

$\therefore$ $\text{I.F.} = e^{\int P dx} = e^{2 \log x} = e^{\log x^2} = x^2$

$\therefore$ The solution of the equation is

$$y \times \text{I.F.} = \int (Q \times I.F.)\, dx$$

$\Rightarrow$ $yx^2 = \int x^3 \log x\, dx$

$\Rightarrow$ $yx^2 = \log x. \frac{x^4}{4} - \int \frac{1}{x} \cdot \frac{x^4}{4}\, dx$

$\Rightarrow$ $yx^2 = \frac{x^4 \log x}{4} - \frac{x^4}{16} + c$

$\therefore$ $y = \frac{1}{4} x^2 \log x - \frac{1}{16} x^2 + \frac{c}{x^2}$.

15. $\mathbf{\frac{dy}{dx} + y = xy^2}$

Sol^n: The given equation is not linear, since $Q = xy^2$ is not function of x only.

So, first we change it into linear equation. Dividing by y^2, we get

$y^{-2} \frac{dy}{dx} + \frac{1}{y} = x$ ..... (i)

Let, $\frac{1}{y} = v$ $\Rightarrow -\frac{1}{y^2} \frac{dy}{dx} = \frac{dv}{dx}$ $\therefore \frac{1}{y^2} \frac{dy}{dx} = -\frac{dv}{dx}$

Hence, equation (i) becomes

$-\frac{dv}{dx} + v = x$ $\Rightarrow \frac{dv}{dx} - (-1)\, v = -x$ ..... (ii)

Which is linear differential equation

$\text{I.F.} = e^{\int -dx} = e^{-x}$

$\therefore$ The solution of the equation is

$$v \times \text{I.F.} = \int (Q \times I.F.)\,dx$$

$\Rightarrow$ $e^{-x}\, v = -\int x\, e^{-x}\, dx$

$\Rightarrow$ $e^{-x}\, v = -[x\,.\,(-e^{-x}) - \int (-e^{-x})\,.\,1\, dx]$

$\Rightarrow$ $e^{-x}\, v = xe^{-x} + e^{-x} + c$

$\Rightarrow$ $v = x + 1 + ce^{x}$

$\Rightarrow$ $\frac{1}{y} = x + 1 + ce^{x}$

$\therefore$ $xy + y + cye^{x} = 1.$

16. $\frac{dy}{dx} + y \tan x = y^3 \sec x$

Sol^n: This equation is not linear. Since $Q = y^3 \sec x$ is not function of x only.

Dividing both sides by y^3, we get

$$y^{-3}\frac{dy}{dx} + y^{-2} \tan x = \sec x \quad \ldots\ldots\ldots\ldots\ldots \text{(i)}$$

Whichis linear differential equation of the form $\frac{dy}{dx} + Py = Q$,

Let, $y^{-2} = v$

$$\Rightarrow -2y^{-3}\frac{dy}{dx} = \frac{dv}{dx} \quad \Rightarrow y^{-3}\frac{dy}{dx} = -\frac{1}{x}\frac{dv}{dx}$$

Hence, equation (i) becomes

$$-\frac{1}{2}\frac{dv}{dx} + v \tan x = \sec x$$

or, $\frac{dv}{dx} + (-2 \tan x)\, v = -2\sec x \quad \ldots\ldots\ldots\ldots\ldots$ (ii)

This is linear equation where $P = -2\tan x$

Now, $\int P dx = -2\int \tan x\, dx = -2\log \sec x = \log \sec^{-2} x = \log \cos^2 x$

$$\text{I.F.} = e^{\int P dx} = e^{\log \cos^2 x} = \cos^2 x$$

$\therefore$ The solution of the equation is

$$v \times \text{I.F.} = \int (Q \times I.F.)\,dx$$

$\Rightarrow$ $v \cos^2 x = -2\int \cos x\, dx = -2 \sin x + c$

$\Rightarrow$ $\frac{\cos^2 x}{y^2} = -2\sin x + c$

$\Rightarrow$ $(c - 2\sin x)\, y^2 = \cos^2 x.$

Hint and solution of MCQ's

1. In the differential equation $\frac{dy}{dx} + Py = Q$

   P and Q both are function in x alone or a constant.

2. $I.F = e^{\int P dx}$

   Since solution of $\frac{dy}{dx} + Py = Q$ is

   $y \times I.F = \int (Q \times I.F)\, dx + c$

   When $Q = 0$

   $y \times I.F = 0 + C$

   $\Rightarrow \quad y \times e^{\int P dx} = C$

   $\Rightarrow \quad y = e^{-\int P dx} C$

3. The I.F of differential equation

   $\frac{dy}{dx} + Py = Q$ is $e^{\int P dx}$

4. Only the differential equation

   $x\frac{dy}{dx} + y = 20$

   $\Rightarrow \quad \frac{dy}{dx} + \frac{1}{x} y = 20$ is of the form $\frac{dy}{dx} + Py = Q$

   Where P is function in x and Q constant.

5. All the differential equations except

   $x\frac{dy}{dx} + y^2 = xy^2$ are of the form $\frac{dy}{dx} + Py = Q$, for P & Q are function in x but not both.

6. $x\frac{dy}{dx} - 2y = x^3$

   $\Rightarrow \quad \frac{dy}{dx} - \frac{2}{x} y = x^2$

   $I.F = e^{\int P dx} = e^{-\int 2/x dx} = e^{-2\log x} = e^{\log x^{-2}} = x^{-2} = \frac{1}{x^2}$

7. $\frac{dy}{dx} + 3y = e^{4x}$

   $I.F \quad = e^{\int P dx} = e^{\int 3 dx} = e^{3x}$

8. $\sin x \frac{dy}{dx} + \cos x.\ y = x \sin x$

   $\Rightarrow \quad \frac{dy}{dx} + \cot x.\ y = x \ \ldots\ldots\ldots$ (i)

   $I.F = e^{\int P dx} = e^{\int \cot x dx} = e^{\log \sin x} = \sin x$

9. $I.F = \frac{1}{x}$

   $\Rightarrow \quad e^{\int P dx} = \frac{1}{x}$

   $\Rightarrow \quad \int P dx = \log \frac{1}{x}$

On differentiation

$$P = \frac{d \log \frac{1}{x}}{d\left(\frac{1}{x}\right)} \times \frac{d\left(\frac{1}{x}\right)}{dx} = \frac{1}{\frac{1}{x}} \times \frac{1}{x^2} = -\frac{1}{x}$$

10. I.F = secx

$\Rightarrow \quad e^{\int P dx} = \sec x$

$\Rightarrow \quad \int P dx = \log \sec x$

On differentiation,

$$P = \frac{d \log(\sec x)}{d(\sec x)} \times \frac{d(\sec x)}{dx}$$

$\Rightarrow \quad P = \frac{1}{\sec x} . \sec x \tan x$

$\Rightarrow \quad P = \tan x.$

11. $\frac{dy}{dx} - y = e^x$ ..... (i) $\qquad [P = -1, Q = e^x]$

I.F. $= e^{\int P dx} = e^{\int -dx} = e^{-x}$

$\therefore$ solution of the equation is

$y \times IF = \int Q \times IF \, dx$

$\Rightarrow \quad ye^{-x} = \int e^{-x} . e^x \, dx$

$\Rightarrow \quad ye^{-x} = \int dx$

$\Rightarrow \quad ye^{-x} = x + c$

$\Rightarrow \quad y = xe^x + ce^x$

12. $x\frac{dy}{dx} + y = x^2$

$\Rightarrow \quad \frac{dy}{dx} + \frac{1}{x} y = x \quad \left[P = \frac{1}{x}, Q = x\right]$

$\therefore \quad I.F = e^{\int P dx} = e^{\int 1/x \, dx} = e^{\log x} = x$

$\therefore$ solution of the equation is

$y \times I.F = \int Q \times I.F. \, dx$

$\Rightarrow \quad yx = \int x^2 dx$

$\Rightarrow \quad yx = \frac{x^3}{3} + c$

$\Rightarrow \quad y = \frac{x^2}{3} + \frac{c}{x}$

13. The solution of $\frac{dy}{dx} + Py = Q$ is

$y \times I.F. = \int Q \times I.F \, dx + c$, which is identical with

$y \sin x = \int x \sin x \, dx + c$

$\therefore \quad I.F = \sin x.$

14. The solution of $\frac{dy}{dx} + Py = Q$ is

$y \times I.F = \int Q \times I.F\ dx + c$

which is identical with

$y\ secx = \int sec^2x\ tanx\ dx + c$

$\Rightarrow \quad ysecx = \int secx\ tanx.\ secx\ dx + c$

$\therefore \quad Q = secx\ tanx$

15. The solution of $\frac{dy}{dx} + Py = Q$ is

$y \times I.F. = \int Q \times I.F\ dx + c$

is identical with

$xy = \frac{x^4}{4} + c$

$\Rightarrow \quad xy = \int x^3 + c$

$\Rightarrow \quad yx = \int x^2.\ xdx + c$

$\therefore \quad Q = x^2$

16. The solution of $\frac{dy}{dx} + Py = Q$ is

$y \times I.F. = \int Q \times I.F\ dx + c$, which is identical with

$y\ sinx = x + c$

i.e. $y\ sinx = \int dx + c$

$\Rightarrow \quad Q \times I.F = 1$

$\Rightarrow \quad Q \times sinx = 1$

$\Rightarrow \quad Q = cosecx$

17. $\frac{dy}{dx} - y = 2xe^x \quad [P = -1, Q = 2xe^x]$

$\therefore \quad I.F = e^{\int Pdx} = e^{\int -dx} = e^{-x}$

$\therefore$ solution of the equation is

$y \times I.F = \int Q \times I.F.\ dx + c$

$\Rightarrow \quad ye^{-x} = \int 2x.\ e^x \times e^{-x}\ dx + c$

$\Rightarrow \quad ye^{-x} = \int 2xdx + c$

$\Rightarrow \quad ye^{-x} = 2.\frac{x^2}{2} + c$

$\Rightarrow \quad y = (x^2 + c)\ e^x$

For $y(0) = 2$ i.e. for $y = 2$ when $x = 0$, we have

$2 = (0^2 + c)\ e^0$

$\Rightarrow \quad 2 = c$

$\therefore \quad y = (x^2 + 2)e^x$